

Dumbbell with viscosity

X26 (2026) — English translation

Part A. Dumbbell with fixed point (2.2 points)

The dumbbell consists of a light rigid rod of length l and two point masses at its ends: m_1 and m_2 ($m_1 > m_2$). There is no gravity field in this part. The dumbbell is fixed at some point on a cylindrical hinge so that it can rotate in the plane containing it. The system is in a viscous medium, so the force of viscous friction acts on the point ends of the dumbbell (and only on them), which is equal to

$$\vec{f}_i = -\gamma \vec{v}_i,$$

где i — номер конца, $\vec{v}_i$ — скорость в лабораторной системе отсчёта, а $\gamma > 0$.

Известно, что момент инерции J of the dumbbell relative to the axis of rotation is minimal among all that can be obtained by fixing the dumbbell in the manner described.

A1	Find the distance x_1 from the end of the dumbbell m_1 to the attachment point.	0.60
A2	Find the moment of inertia J .	0.20

At the moment of time $t = 0$ the dumbbell was at rest, and a force F , constant in magnitude, was applied to the heavy end m_1 , the direction of which is always perpendicular to the arm of the dumbbell.

A3	Calculate the angular acceleration ε of the dumbbell at time $t = 0$.	0.30
A4	Calculate the steady-state angular velocity ω_{lim} of the dumbbell rotation after a long time.	0.60
A5	Find the moment of time τ at which the angular velocity is equal to $\omega_{\text{lim}}/2$.	0.50

Part B. Free Dumbbell (1.4 points)

At this point, the gravity field and friction force are absent. Now the dumbbell is not secured.

Let us introduce a rectangular Cartesian coordinate system Oxy . At the moment of time $t = 0$ the dumbbell was at rest along the axis Ox and a force F , constant in magnitude, was applied to its heavy end m_1 , the direction of which is always perpendicular to the arm of the dumbbell. At the initial moment, the force is co-directed with the Oy axis.

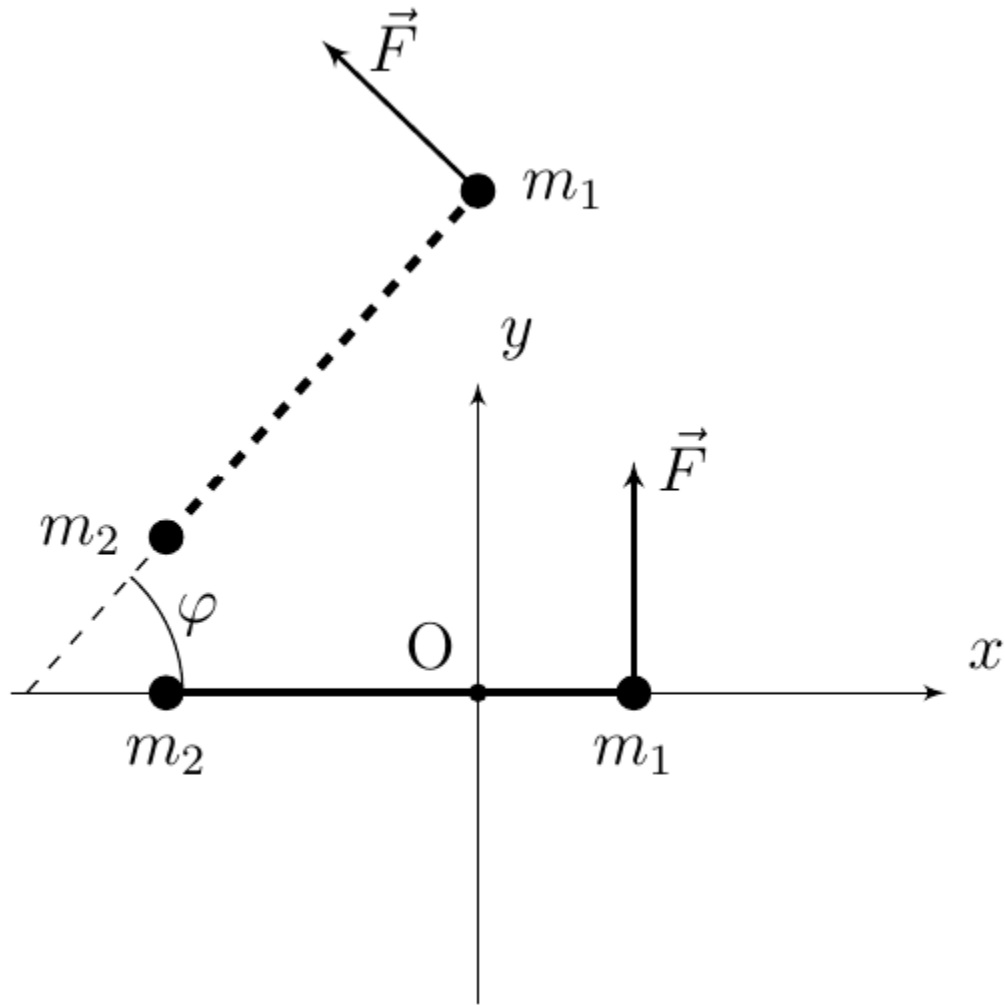


Fig. 1.

B1	Calculate the angular acceleration ε of the dumbbell at time $t = 0$.	0.20
B2	How long will T_n take for the n th rotation of the dumbbell?	0.20
B3	After a long time, the speed of the dumbbell's center of mass becomes almost constant. Find the vector $\vec{u}$ of the steady-state velocity of the center of mass. In response, write down the x - and y -components of the $\vec{u}$ velocity.	1.00

You may need the following Fresnel integrals:

$$\int_0^{\infty} \sin(x^2) dx = \int_0^{\infty} \cos(x^2) dx = \sqrt{\frac{\pi}{8}}.$$

Part C. Relaxation in a viscous medium (1.8 points)

Now the dumbbell is not fixed, it is in a viscous medium in the gravitational field, and it is also acted upon by a constant Archimedes force, the point of application of which is in the middle of the dumbbell. Its modulus is equal to the force of gravity. Initially, the dumbbell rests in a vertical position.

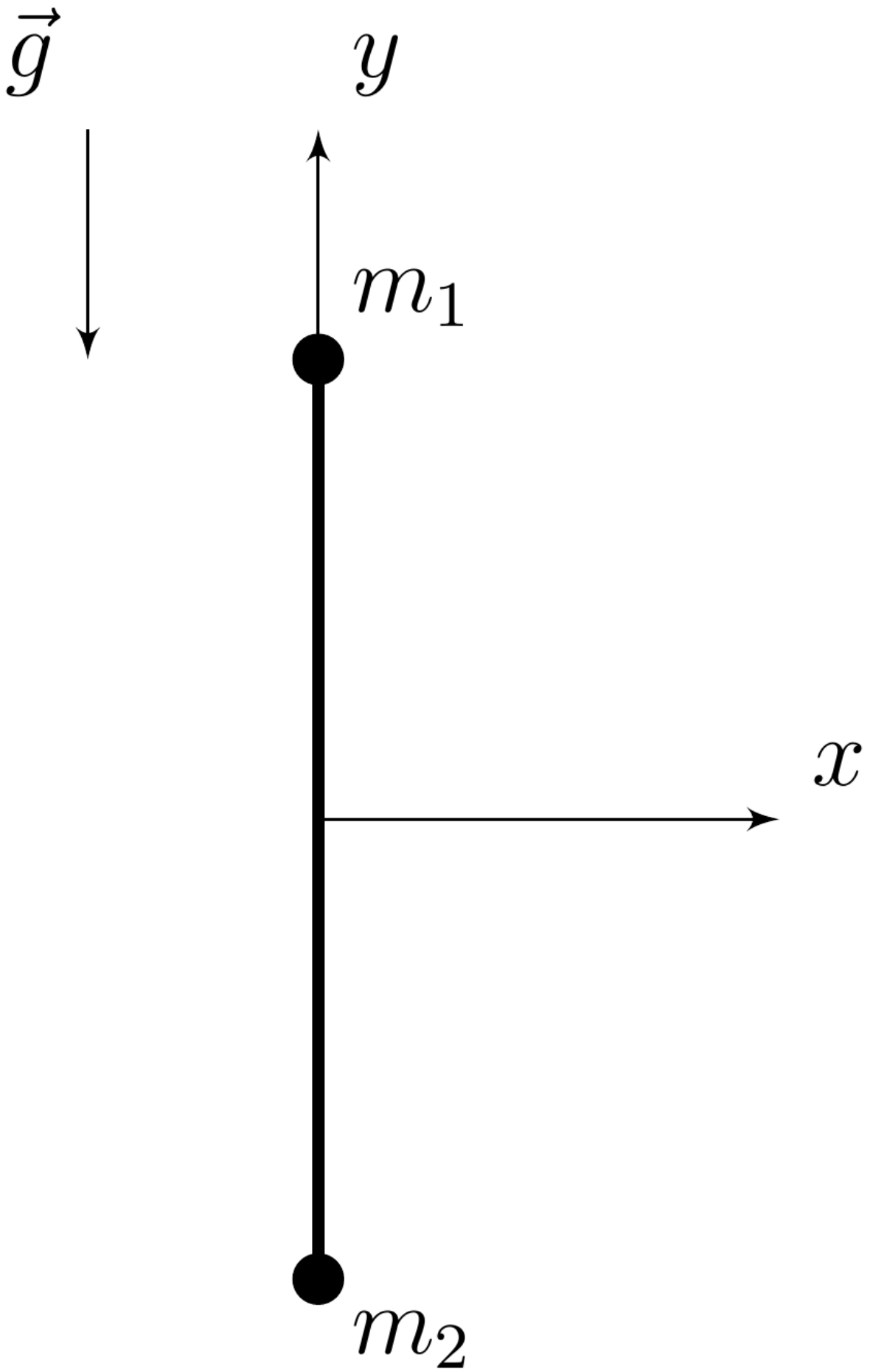


Fig. 2.

In this part of the problem and further, for convenience, when solving and expressing the answer, you can use the following notation: by $\vec{\Delta}$ we denote the vector, the beginning of which is located at the center of mass of the dumbbell, and the end - in its middle; $\Delta = |\vec{\Delta}|$.

The dumbbell is slightly tilted from its equilibrium position.

C1	Determine the final displacement vector $\vec{s}$ of the center of mass of the dumbbell after it stops. The answer must be presented in the form of components in the coordinate system shown in this figure.	1.50
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C2	Determine how much heat was released during the entire movement of the dumbbell.	0.30
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Part D. Vibrations in a viscous medium (1.9 points)

Now, under the conditions of the previous point, the dumbbell is in a stable vertical equilibrium position: the heavy end m_1 is strictly under the light end m_2 .

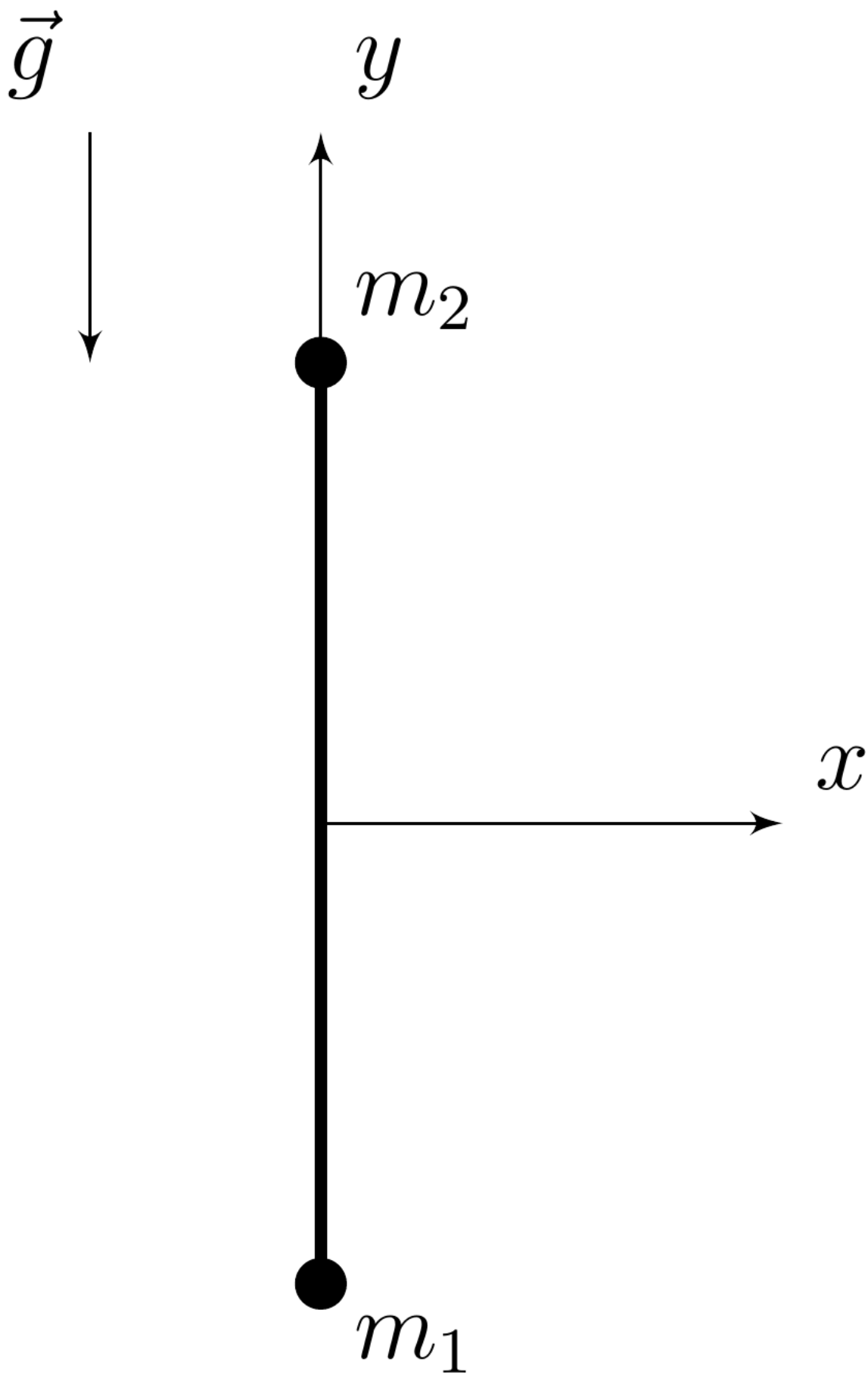


Fig. 3.

The dumbbell is deflected from the vertical by an angle $\varphi_0 \ll 1$, and then released without initial speed. Consider the energy losses caused by viscous friction to be small.

D1 At an arbitrary moment in time, write down the total moment $\vec{M}_{\text{out}}$ of the external forces relative to the center of mass of the dumbbell. Express it in terms of the center of mass velocity $\vec{v}$, angular velocity $\vec{\omega}$, $\vec{\Delta}$, $\vec{g}$ and system parameters. **0.60**

D2 Show that under the conditions of small damping and $\varphi_0 \ll 1$ in the equation of rotational motion we can neglect the moment caused by the translational motion of the center of mass. **0.20**

Even if you did not complete the previous step, the approximation introduced in it can be used further.

D3 Write down the equation of rotational motion. Taking into account the smallness of the damping and the condition $\varphi_0 \ll 1$, obtain the equation of damped oscillations of the angle formed by a dumbbell with a vertical: **0.50**

$$\ddot{\varphi} + A\dot{\varphi} + B\varphi = 0.$$

Определите коэффициенты A и B .

D4 Determine the period T of the dumbbell oscillations. **0.20**

D5 Estimate the characteristic time τ of damping of the dumbbell oscillations. **0.20**

D6 Determine the quality factor Q of such an oscillatory system. **0.20**

Part E. Regular dumbbell precession (0.8 points)

Now let the dumbbell rod be a homogeneous cylinder with radius R , length l and mass m . In this section we will consider the precession of such a “top” under the influence of its own gravity. From now on until the end of the problem, assume $m_1 = m_2 = 0$.

E1 Calculate the moment of inertia I of the dumbbell rod about its axis. **0.10**

You can always use the notation I when solving and expressing your answers.

One of the ends of the dumbbell is fixed to the table surface on a spherical joint without friction. The dumbbell rotates around its axis with a very high angular velocity ω , so we can further assume that the total angular momentum $\vec{L}$ of the dumbbell is directed along the vector $\vec{\omega}$. In addition to the reaction force of the support, the force of gravity also acts on the dumbbell, the moment of which causes a movement called gyroscopic (or regular precession): the center of mass of the dumbbell rotates around the vertical passing through the spherical joint with a certain steady angular velocity Ω .

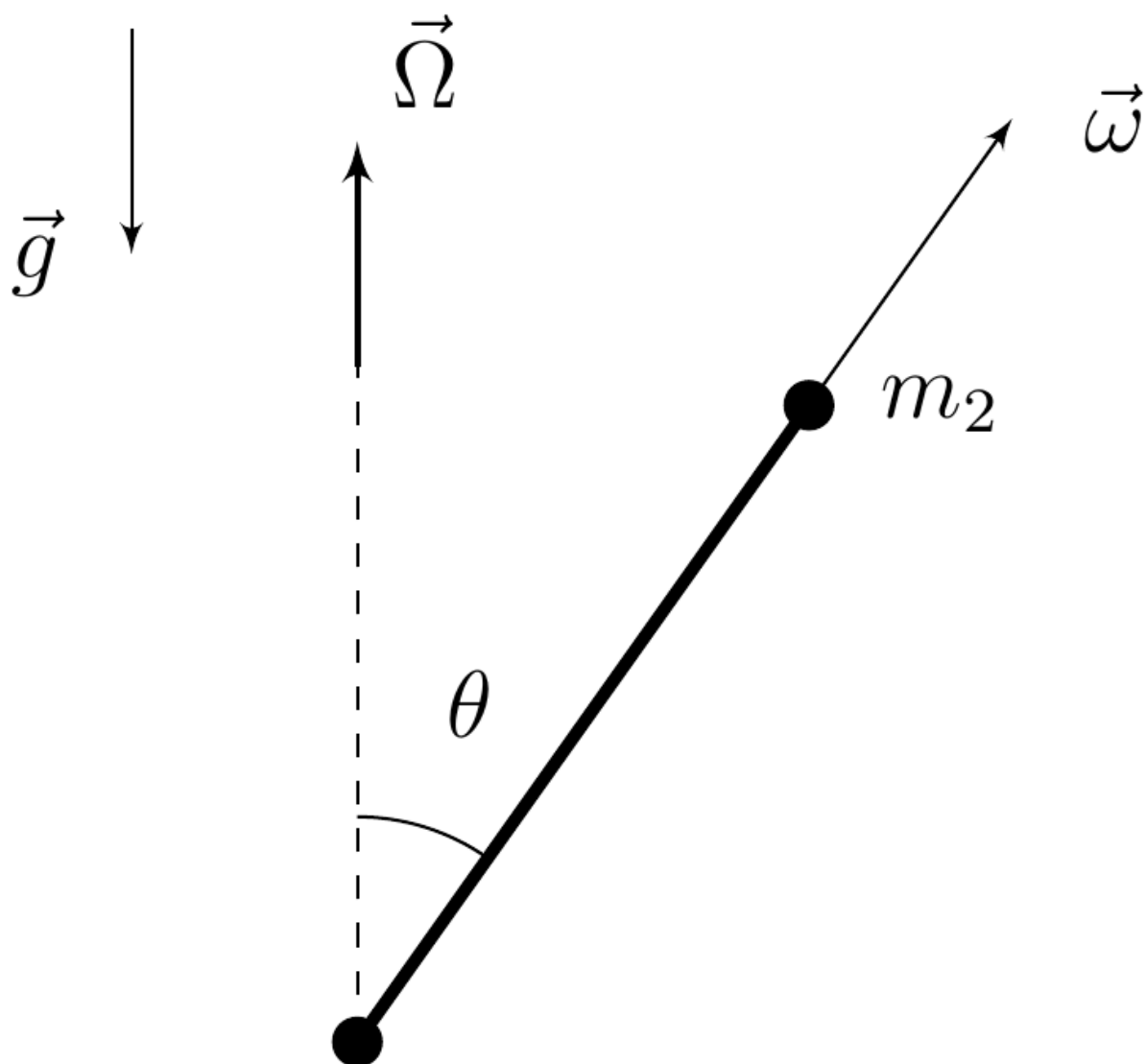


Fig. 4.

Let the dumbbell perform the described rotation in a uniform gravity field $\vec{g}$, moreover, the angle of the dumbbell shoulder with the vertical is constant and equal to θ_0 .

E2	Write down the law of change in the angular momentum of a dumbbell relative to its attachment point.	0.20
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E3	Calculate the angular velocity $\vec{\Omega}$ of the regular precession of the dumbbell.	0.50
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Part F. Precession in the presence of viscous friction (1.9 points)

Now let the dumbbell perform gyroscopic motion in the presence of viscous friction. On the one hand, the non-supported end of the dumbbell m_2 is subject to the force of viscous friction associated with its precessional movement around the vertical. On the other hand, there is a moment of viscous friction forces caused by the rotation of the dumbbell around its own axis of symmetry. It is equal to $\vec{M}_{\text{fric}} = -\alpha\vec{\omega}$, $\alpha > 0$. Further, consider the following approximations always correct: $\omega \gg \Omega$ and $\dot{\theta} \ll \frac{mg}{\gamma l}$.

F1	Calculate the dependence of the dumbbell angular momentum modulus on time.	0.30
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F2	Derive a differential equation describing the change in the angle of the dumbbell with the vertical over time.	0.60
F3	Let the initial angle be θ_0 . What is the rate of change of the angle at the initial moment?	0.20
F4	With the initial condition from the previous paragraph, obtain an explicit dependence $\theta(t)$.	0.80

Source: <https://pho.rs/p/4800>